INCH-POUND

MIL-DTL-6458E
14 February 2003
SUPERSEDING
MIL-DTL-6458D
15 December 1995

DETAIL SPECIFICATION

CHAIN ASSEMBLIES, SINGLE LEG, AIRCRAFT CARGO TIE DOWN

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of flight safety critical aircraft part (FSCAP) cargo tie down chain assemblies. The type I chain assembly is to be used in conjunction with MIL-DTL-25959, type I, styles A and B cargo tie down adjusters. The type II chain assembly is to be used in conjunction with MIL-DTL-25959, type II, styles A and B cargo tie down adjusters.

1.2 Classification. Chain assemblies covered by this specification are of the following types:

Type I - 10,000 pound capacity (see 6.3)
Type II - 25,000 pound capacity (see 6.3)

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data that may be of use in improving this document should be addressed to: Defense Supply Center Richmond, ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1670

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2.2 Government documents.

2.2.1 Standards. The following standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS), and the supplement thereto, cited in the solicitation (see 6.3).

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-810	- Environmental Engineering Considerations and Laboratory Tests.
MIL-STD-889	- Dissimilar Metals.

(Unless otherwise indicated, copies of the above standards are available from the Standardization Documents Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094. Electronic copies of military standards may be obtained from http://astimage.daps.dla.mil/quicksearch/.)

2.3 Non-government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of documents that are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DoDISS are the issues of documents cited in the solicitation (see 6.3).

ASTM INTERNATIONAL

ASTM A 304	- Standard Specification for Carbon and Alloy Steel Bars Subject to End-Quench Hardenability Requirements (DoD adopted).
ASTM A 322	- Standard Specification for Steel Bars, Alloy, Standard Grades (DoD adopted).
ASTM A 331	- Standard Specification for Steel Bars, Alloy, Cold-Finished (DoD adopted).

(Application for copies should be addressed to the ASTM International, 100 Barr Harbor Drive, P.O. Box 700C, West Conshohocken, PA 19428-2959. Electronic copies may be obtained from http://www.astm.org/.)

AMERICAN WELDING SOCIETY

AWS B1.11	- Guide for the Visual Examination of Welds.

(Application for copies should be addressed to the American Welding Society, 550 N.W. LeJeune Road, P.O. Box 351040, Miami, FL 33135. Electronic copies may be obtained from http://www.aws.org/.)

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SOCIETY OF AUTOMOTIVE ENGINEERS, INC.

SAE-AMS-H-6875 - Heat Treatment of Steel Raw Materials (DoD adopted).

(Application for copies should be addressed to Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096-0001. Electronic copies may be obtained from http://www.sae.org/.)

2.4 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification takes precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.3), a sample shall be subjected to first article inspection in accordance with 4.3.

3.2 Materials.

3.2.1 Metals. Metals shall be suitably treated to resist corrosion due to fuels, salt fog (spray), or atmospheric conditions likely to be met in storage or normal service. The chain shall be made of alloy steel conforming to ASTM A 304, ASTM A 322, ASTM A 331, or equivalent, and heat treated to produce a chain of high tensile strength with resistance to impact loads.

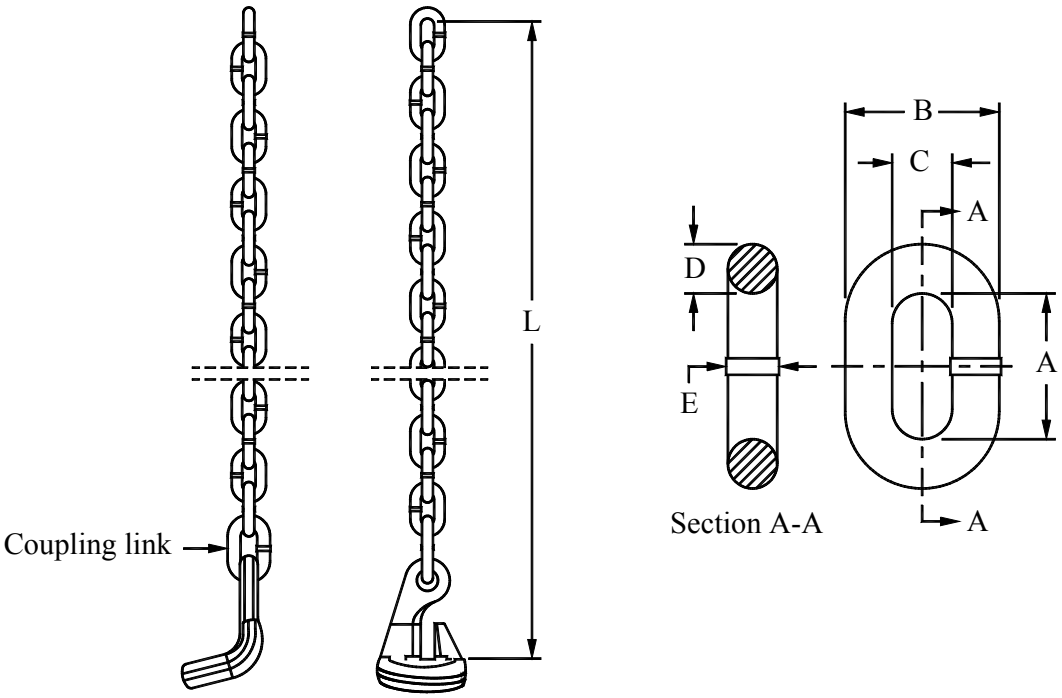
3.2.2 Heat treatment. Heat treatment of steel parts shall be in accordance with SAE-AMS-H-6875.

3.2.3 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.3 Chain assembly. The chain assembly shall be of the welded-link type consisting of a length of chain, a coupling link, and a hook constructed so that it forms a single unit without detachable parts. The weld on each link shall not interfere with attaching that link to or removing it from the adjuster tumble block (i.e. no sharp edges extending beyond links). Each link shall be capable of being attached to and removed from the adjuster tumble block, with a load applied, without the use of special tools. The chain assembly shall meet the strength, weight, and dimensional requirements specified in figure 1.

3.4 Hook. The hook shall be a grab type with dimensions as shown in figure 2 and shall engage chain links of the size specified in figure 1. The hook shall be designed to permit fastening of one chain assembly to another of the same type to form a single branch system of two chain assemblies for extension purposes. The hooks shall meet the design strength requirements of the chain assemblies and shall not reduce the strength of the chain assembly into which it has been attached. The hook must fully engage the attaching chain link with the chain link in contact with radius R (see figure 2) and without interference from adjoining links.

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Chain assembly			
Type	Design Strength (lbs.)	Max. Weight (lbs.)	Min. Length (inches)
I	14,100	8	108
II	35,250	24	108

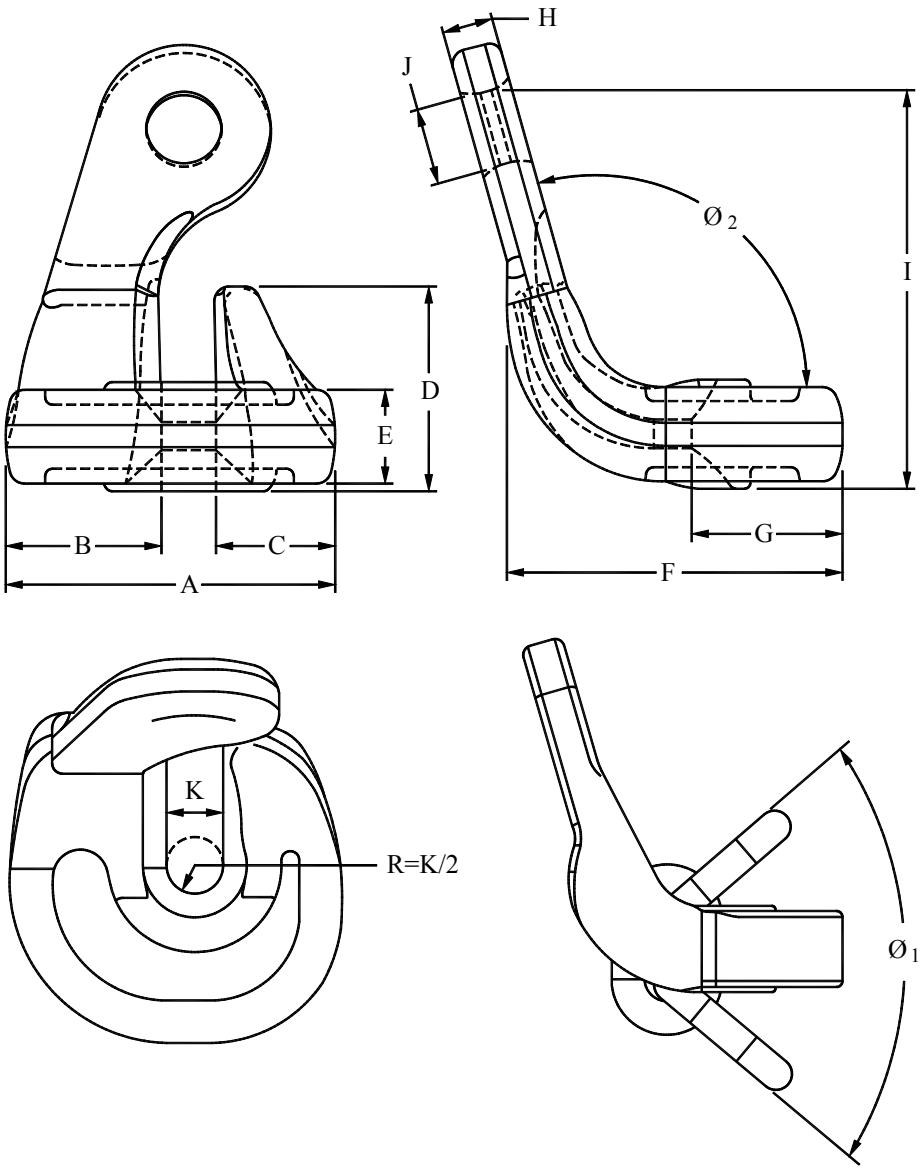
Coupling link dimensions (inches)														
Type	Inside length - A			Width				Bar size diameter - D			Weld diameter - E		
				Outside - B		Inside - C							
	Nominal	Min.	Max.	Min.	Max.	Min.	Max.	Nominal	Min.	Max.	Nominal	Min.	Max.
I	1.020	0.961	1.079	–	1.220	0.453	–	0.354	0.346	0.362	–	–	0.394
II	1.496	1.488	1.503	–	1.889	0.669	–	0.519	0.511	0.519	–	–	0.560

Chain link dimensions (inches)													
Type	Inside length - A			Width				Bar size diameter - D			Weld diameter - E		
				Outside - B		Inside - C							
	Nominal	Min.	Max.	Min.	Max.	Min.	Max.	Nominal	Min.	Max.	Nominal	Min.	Max.
I	0.923	0.923	0.941	–	0.975	0.374	0.394	0.281	0.275	0.281	–	–	0.307
II	1.456	1.417	1.503	–	1.515	0.535	–	0.472	0.464	0.472	–	–	0.502

Note: The "B" dimension includes the weld dimension "E".

Figure 1. Chain assembly – strength, weight, and dimensional requirements.

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Type	A	B	C	D	E	F	G	H	I	J	K	R	$\varnothing_1$	$\varnothing_2$
I	2.266	1.047	0.844	1.359	0.650	2.250	1.031	0.359	2.657	0.515	0.375	0.188	50°	105°
II	3.425	1.368	1.368	1.968	1.102	3.780	1.673	0.590	3.661	0.866	0.610	0.305	75°	105°

Tolerances: A maximum; B, C, D, E, F, G, and I approximate; H + 0.04/- 0 inch; J and K + 0.028/- 0.028; $\varnothing_1$ and $\varnothing_2$ approximate.

Figure 2. Grab hook dimensional requirements.

3.5 Performance.

3.5.1 Proof load. When tested as specified (see 4.5.2), the type I chain assembly shall be capable of withstanding proof loads of 10,000 pounds and the type II chain assembly shall be capable of withstanding proof loads of 25,000 pounds. There shall be no evidence of failure, permanent deformation, or subsequent malfunction of parts.

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3.5.2 Ultimate load. When tested as specified (see 4.5.3), the type I chain assembly shall be capable of withstanding ultimate loads of 14,100 pounds and the type II chain assembly shall be capable of withstanding ultimate loads of 35,250 pounds. Deformation, without rupture or visible cracks, will be permitted provided the chain remains flexible after removing the applied load.

3.5.3 Environmental characteristics. The chain assembly shall be fully operational and show no evidence of corrosion when subjected to the salt fog (spray) testing (see 4.5.5).

3.6 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable.

3.7 Finish. All steel parts with the exception of corrosion resistant steel shall be treated with a metallic-type coating that will provide a cathodic protective coating to the base metal. This protective coating shall not crack, peel, or flake during normal usage and shall withstand any adverse conditions to which the chain assembly will be subjected in accordance with this specification. The protective coating shall in no way prevent compliance with the performance requirements of this specification.

3.8 Identification of product. The chain assembly shall be identified by means of a metal tag permanently attached to the end link of the chain. The identification tag shall be marked in accordance with MIL-STD-130 and contain as a minimum the national stock number (NSN), contract number, and contractor's name or manufacturer's code. The required marking shall be applied to a metal tag that is attached to the first link by a metal ring or link. This tag shall remain attached for the life of the chain. The first link is at the end of the chain furthest away from the master link attaching the chain to the hook.

3.9 Workmanship. The chain assembly shall be free from injurious imperfections that may impair appearance and serviceability. The welded chain shall be free from all surface discontinuities in accordance with AWS B1.11. Dimensions and tolerances not specified shall be consistent with standard industry practice.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Conformance inspection (see 4.4).

4.2 Test conditions. Unless otherwise specified, all tests required by this specification shall be made at standard ambient conditions in accordance with MIL-STD-810.

4.3 First article inspection. The first article inspection shall consist of all tests specified in section 4.5.

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4.3.1 First article sample. The first article test sample shall consist of two chain assemblies of the type required that are representative of the production chain assemblies. The test samples shall undergo all of the tests specified in section 4.5 at a laboratory designated by the plant under the supervision of the procuring activity.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-STD-1916 and shall consist of the individual tests (see 4.4.1) for each unit produced, as well as the periodic sampling tests (see 4.4.2) if required by the procuring activity (see 6.3).

a. Individual tests.
b. Periodic sampling tests.

4.4.1 Individual tests. Each chain assembly shall be subjected to the examination of product and proof load test (see 4.5.1 and 4.5.2).

4.4.2 Periodic sampling tests. The periodic sampling tests shall be in addition to the individual tests and shall consist of the tests specified in 4.5.3 and 4.5.4. The manufacturer shall select the type of sampling plan (i.e. attribute, variable, or continuous) in accordance with MIL-STD-1916. The sample size shall be selected in accordance with verification level I of MIL-STD-1916.

4.4.3 Rejection and retest. Failure of any item being tested and inspected will postpone final acceptance until the cause of rejection has been investigated and corrected for the entire contract quantity. Other than marking and minor workmanship issues as determined by the government, items that have been tested and failed shall be discarded and not be reworked or delivered as part of any end item lot. If a new production lot is required, the lot will be tested using a tightened sample size in accordance with verification level II of MIL-STD-1916.

4.5 Test methods.

4.5.1 Examination of product. Each chain assembly shall be inspected to determine compliance with the requirements specified herein with respect to material, dimensions, weight, protective coating, weld quality, marking, and workmanship.

4.5.2 Proof load test. Each chain assembly shall be tested with the static load specified in 3.5.1 applied to the grab hook and chain links for 30 seconds without permanent deformation or damage when wrapped around a 6-inch diameter bar with the hook fastened into a link maintaining an included angle of 30 ± 5 degrees.

4.5.3 Ultimate load test. Each sample chain assembly shall be subjected to the design strength test in accordance with the procedure in 4.5.2, except that the static load requirements shall be as specified in 3.5.2.

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4.5.4 Hook test. Each sample chain assembly of the same type shall be joined as specified in 3.4 and subjected to a load equivalent to the design strength. The hook and adjoining links shall be examined under full load for positive engagement. There shall be no evidence of slippage or impending failure of the hook, coupling link, or chain.

4.5.5 Salt fog (spray) test. The first article sample chain assembly shall be subjected to salt fog (spray) test in accordance with MIL-STD-810. After completion of the test, the chain assembly shall show no evidence of rust or corrosion (see 3.5.3).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or acquisition order (see 6.3). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the inventory control point's packaging activity within the military department or defense agency, or within the military department's system command. Packaging data retrieval is available from the managing military department's or defense agency's automated packaging files, CD-ROM products, or by contracting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The chain assemblies covered by this specification are to be used in conjunction with MIL-DTL-25959 cargo tie down adjusters for securing vehicles and equipment in cargo type aircraft during flight.

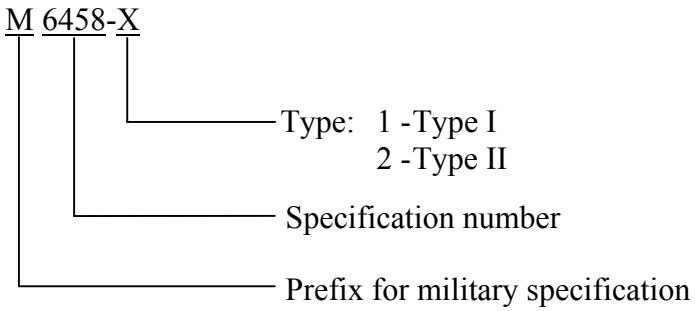
6.2 Acquisition process. This specification is for a flight safety critical aircraft part (FSCAP) and the acquisition process should comply with DoD Materiel Management Regulation DoD 4140.1-R.

6.3 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type of chain assembly required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).
d. When first article is required (see 3.1).
e. When periodic sampling tests are required (see 4.4).
f. Packaging required (see 5.1).

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6.4 Part or identifying number (PIN). The PIN to be used for chain assemblies acquired to this specification is created as follows:



6.5 Subject term (key word) listing.

coupling link
hook
steel alloy

6.6 Changes from the previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - AV
Navy - AS
Air Force - 99

Preparing Activity:
DLA - GS1

(Project 1670-1003)

Review Activities:
Air Force - 71

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-DTL-6458E	**2. DOCUMENT DATE** (YYYYMMDD) 20030214

3. DOCUMENT TITLE
CHAIN ASSEMBLIES, SINGLE LEG, AIRCRAFT CARGO TIE DOWN

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) (1) Commercial (2) DSN *(if applicable)*	**7. DATE SUBMITTED** (YYYYMMDD)

8. PREPARING ACTIVITY

a. NAME Defense Supply Center Richmond	b. TELEPHONE (*Include Area Code*) Commercial: (804) 279-5019 DSN: 695-5019
c. ADDRESS *(Include Zip Code)* ATTN: DSCR-VBD (C. Hammond) 8000 Jefferson Davis Highway Richmond, VA 23297-5610	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** DEFENSE STANDARDIZATION PROGRAM OFFICE (DLSC-LM) 8725 John J. Kingman Road, Suite 2533 Fort Belvoir, VA 22060-6221 Commercial: (703) 767-6888 DSN: 427-6888

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